UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2009 question paper for the guidance of teachers

5070 CHEMISTRY

5070/03 Paper 3 (Practical Test), maximum raw mark 40

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UNIVERSITY of CAMBRIDGE
International Examinations

1 (a)

Test		Notes
Test **1**		
(a) White ppt	(1)	Ppt must be white
(b) Disappears	(1)	
Test **2**		
(a) White ppt	(1)	Ppt must be white
(b) Remains	(1)	

Formula of impurity in **P** is NaC*l* or C*l*$^-$ (Test 2 must be correct) (1) [5]

(b) Titration

Accuracy

For each of the two best titres give:
4 marks for a value within 0.2 cm^3 of supervisor
2 marks for a value within 0.3 cm^3 of supervisor
1 mark for a value within 0.4 cm^3 of supervisor (8)

Concordance

Give:
3 marks if all the ticked values are within 0.2 cm^3
2 marks if all the ticked values are within 0.3 cm^3
1 mark if all the ticked values are within 0.4 cm^3 (3)

Average

Give 1 mark if the candidate calculates a correct average (error not greater than 0.05) of all his ticked values.

Assuming a 25 cm^3 pipette and a titre of 24.8 cm^3. (1) [12]

(c) moles of sodium carbonate in 1.00 dm^3 of **P**

$= \frac{24.8 \times 0.1}{25 \times 2}$ (1)

= 0.0496 (correct to 0.0001) (1) [2]

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(d) mass of sodium carbonate in 1 dm^3 of solution **P**.

= 0.0496 × 106

= 5.26 g [1]

(e) percentage by mass of sodium carbonate in impure sample. [1]

$$= \frac{5.26 \times 100}{6.00}$$

= 87.7%

[Total: 21]

2 **R** is zinc sulfate **S** is iron(II) sulfate **T** is lead(II) nitrate

Test		Notes
General points For ppt allow solid, suspension, powder For gases Name of gas requires test to be at least partially correct. Effervesces = Bubbles = gas vigorously evolved (but not just gas evolved) Solutions Colourless not equivalent to clear, clear not equivalent to colourless		
Solution **R**		
Test **1** **(a)** White ppt **(b)** Soluble in excess Colourless solution	 (1) (1) (1)	 Ppt must be white
Test **2** **(a)** White ppt **(b)** Soluble in excess Colourless solution **(c)** No reaction	 (1) (1) (1) (1)	 Ppt must be white
Test **3** No reaction/purple colour remains	 (1)	 Any indication of a reaction = 0

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Solution **S**		
Test **1**		
(a) Green ppt	(1)	
(b) Insoluble in excess	(1)	
Test **2**		
(a) Green ppt	(1)	
(b) Insoluble in excess	(1)	
(c) Red-brown ppt	(1)	Accept brown but not orange
Effervescence	(1)	
Gas relights glowing splint	(1)	Test for gas here or in Test **2 (c)** for **T**
Oxygen	(1)	Partially correct test allow mark for oxygen
Test **3**		
Decolourised	(1)	Colourless solution formed

Solution **T**		
Test **1**		
(a) White ppt	(1)	
(b) Insoluble in excess	(1)	
Test **2**		
(a) White ppt	(1)	
(b) Soluble in excess	(1)	
Colourless solution	(1)	
(c) Black/brown ppt	(1)	
Effervescence	(1)	
Oxygen	(1)	
Test **3**		
White ppt	(1)	
No reaction/purple colour remains	(1)	

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Conclusion

R is Zn^{2+} (White ppt must disappear in Tests 1 & 2) (1)

S is Fe^{2+} (Green ppt in 1 or 2) (1)

Any candidate obtaining 25 or more marks scores the maximum on question 2 i.e. 19. Thereafter scores are awarded according to the following table

Marks	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6
Score	19	19	18	17	17	16	15	14	14	13	12	11	11	10	9	8	8	7	6	5	5

Marks	5	4	3	2	1
Score	4	3	2	2	1

[Total: max 19]

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